

## Graduate studies in Elementary Particle Theory at the Department of Physics

Dear applicant,

Thank you for your interest in the Zürich Ph.D. programs in physics. Eligible applicants must have a Master's or Diploma degree or an equivalent at the start of the program, which is within 6 months of acceptance by the admission committee.

**Application form.** Please complete the form in English (save it as *Familyname-Firstname.doc*) and submit it through Academic Jobs Online. Additionally, please include a **scanned** version of your Diploma or Master's certificate and grade transcripts. Also include scanned official GRE and TOEFL documents, if available.

**Letters of recommendation.** In addition to the completed application form, your CV and transcript record we request letters of recommendation from two scientists who are familiar with your previous academic work and who can judge your potential as a Ph.D. student. **Send the recommendation form to your referees. Note that we will be unable to process your application without the letters of recommendation!**

Please list the names and email addresses of your referees here:

- 1.) **Francesco Sannino**, sannino@mac.com
- 2.) **Sofie Marie Koksang**, koksang@cp3.sdu.dk

**Assessment and selection procedure.** Applications are reviewed by an admission committee composed of faculty members of the primary program chosen by the applicant, they will be assessed on the basis of their application and letters of recommendation. Short-listed candidates may be invited for an interview in Zürich. We will reimburse your travel expenses and provide free accommodation in Zürich. Selection for admission will be based on the perceived ability and potential of the student. Offers of admission to the Ph.D. program will be made soon after the interview.

Graduate Studies at the Department of Physics

## APPLICATION FORM FOR PhD STUDIES

**Date: 01/12/2025**

**Please indicate which program(s) you are applying to:**

☒ Theoretical Particle Physics

|                                                                                     |                                       |                                  |
|-------------------------------------------------------------------------------------|---------------------------------------|----------------------------------|
| <input type="checkbox"/> <b>Mr</b><br><input checked="" type="checkbox"/> <b>Ms</b> | Family name<br><b>Wagner</b>          | First name<br><b>Sophie Lund</b> |
| Street<br><b>Falen 12 4. th.</b>                                                    |                                       |                                  |
| Postal code, City<br><b>5000</b>                                                    |                                       |                                  |
| Country<br><b>Denmark</b>                                                           | Email<br><b>wsophielund@gmail.com</b> | Phone<br><b>+45 42187737</b>     |
| Date of birth<br><b>14/12/1999</b>                                                  | Nationality<br><b>Danish</b>          |                                  |

## EDUCATION

**Highest degree** (completed or expected; e.g. Master's, Diploma, DEA)

|                                                             |                                                                                                |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| University/College<br><b>University of Southern Denmark</b> | Dates (from-to)<br><b>2024-2026</b>                                                            |
| Degree* (e.g. Master, Diplom, DEA)<br><b>Master</b>         | Grade# (if known) (on a scale of <b>12</b> (best) to <b>-2</b> (worst))<br><b>12 (current)</b> |

|                                                                             |                                                                                                    |                                          |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------|
| Date (completed or expected)<br><b>01/07/2026</b>                           | Major field of study<br><b>Physics</b>                                                             | Minor field of study<br><b>Astronomy</b> |
| Subject of thesis<br><b>Novel Principles for Grand Unification Theories</b> | Name and address of thesis supervisor (if applicable)<br><b>Francesco Sannino, sannino@mac.com</b> |                                          |

\*If you already have obtained your highest degree, please fax or send a photocopy of the certificate

**Previous degree(s)** (e.g. Bachelor, Vordiplom, Maîtrise)

|                                                                                              |                                                                                              |                                     |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------|
| University/College<br><b>Southern University of Denmark</b>                                  |                                                                                              | Dates (from-to)<br><b>2021-2024</b> |
| Degree<br><b>Bachelor</b>                                                                    | Grade# (on a scale of <b>12</b> (best) to <b>-2</b> (worst))<br><b>11.53</b>                 |                                     |
| Date<br><b>2024</b>                                                                          | Major field of study<br><b>Physics</b>                                                       | Minor field of study                |
| Field of work<br><b>Theoretical Physics, with a thesis regarding inhomogeneous cosmology</b> | Name of project supervisor (if applicable)<br><b>Sofie Marie Koksang, koksang@cp3.sdu.dk</b> |                                     |

#Give final or average grade here, list grades in detail on page 5

## EDUCATION AND TRAINING

Please give a brief summary of your education and training at the university and/or college etc. If your education has been interrupted, or if you have already obtained other Diplomas, please specify the occupation(s) you have pursued in the interim. Do not exceed this page.

I started my university education in 2021 at the University of Southern Denmark with a bachelor's degree in physics. Throughout the three years, I completed multiple courses in physics, mathematics, and astronomy, with a primary focus on theoretical physics. After this degree, my future path was clear to me; I therefore continued with a master's degree in physics with a specialisation in fundamental physics in 2024 at the same university. However, to challenge myself and gain further experience in other research groups and environments, I went on exchange to the University of Calgary, Canada. Here I furthered my knowledge within theoretical physics, especially within the quantum world. Additionally, I chose to improve my scientific communication skills through specific courses.

## **COURSE WORK**

Please list the courses you have taken and the grades obtained. Include all university level or other relevant courses. If you have passed final university examinations, please give details. Please indicate

the best and worst grade in the scale used by your university. If you have completed non-graded laboratory courses, please give name(s) of supervisor(s). Do not exceed this page.

The grade scale best to worst is: 12, 10, 7, 4, 02, 00, -2

**Courses in Master's: (Current Average 12)**

Quantum Field Theory, 10 ects, Grade: 12.

General relativity and cosmology, 5 ects, Grade: 12.

Advanced statistical physics, 10 ects, Grade: 12.

Individual Study Activity in Physics (Anomalies in Quantum Field Theory), 5 ects, Grade: Passed.

Observational astronomy, 5 ects, Grade: Passed.

Oral Science Communication, 5 ects, Grade: Passed.

Research activities in experimental physics, 5 ects, Grade: Passed.

**Courses on exchange semester (Calgary University, Canada):**

Quantum optics (Transferred), 5 ects, Grade: Passed.

Graduate seminar (Transferred), 5 ects, Grade: Passed.

Electrodynamics (Transferred), 5 ects, Grade: Passed.

**Courses in Bachelor's: ( Average: 11.53 )**

Mechanics and thermodynamics, 10 ects, Grade: 12.

Calculus, 10 ects, Grade: 12.

Electronics, 5 ects, Grade: 12.

Experimental Design and Modelling, 5 ects, Grade: 12.

Electromagnetism and Optics, 10 ects, Grade: 10.

First Year Project, 10 ects, Grade: 12.

Digital Electronics and Programming, 10 ects, Grade: 12.

Quantum mechanics I, 5 ects, Grade: 12.

Advanced Mechanics and Relativity Theory, 10 ects, Grade: 7.

Linear algebra, 5 ects, Grade: 12.

Numerical physics, 5 ects, Grade: 12.

Statistics, 5 ects, Grade: 12.

Quantum mechanics II, 5 ects, Grade: 12.

Statistical Physics, 5 ects, Grade: 12.

Electrodynamics, 5 ects, Grade: 12.

Experimental physics and semiconductors, 5 ects, Grade: 12.

Fundamental Astronomy, 5 ects, Grade: 12.

Solid state physics, 10 ects, Grade: 12.

Astrophysics and Introduction to Cosmology, 5 ects, Grade: 12.

Particle physics, 5 ects, Grade: 12.

Partial Differential Equations and complex analysis, 5 ects, Grade: 12.

Quantum Physics, 10 ects, Grade: 10

Introduction to Python, machine learning and data handling, 5 ects, Grade: Passed.

Individual Study Activity in Physics (Dark matter and neural networks), 5 ects, Grade: Passed.

Galactic dynamics and dark matter, 5 ects, Grade: 10.

Bachelor's project in Physics (Cosmic backreaction), 15 ects, Grade: 12.

## RESEARCH EXPERIENCE

Please give the title and a short description of any research project carried out during your studies (diploma thesis, M.Sc. thesis etc). State the rationale and goals of the research, and indicate your own

contributions to the results. In all cases, state the number of months of research involved. List publications, if any. Do not exceed this page.

My research experience spans fundamental particle physics, cosmology, and computational methods. Currently, I am working on my master's thesis, which focuses on gaining a deeper understanding of fundamental interactions and of beyond Standard Model physics. In my research, I explore the possibility of underlying symmetries at higher energies, where the strong, weak, and electromagnetic interactions may merge into a single gauge structure, as suggested by GUTs. Rather than proposing a specific gauge group, I study a general landscape of high-energy GUTs and identify preferred models by minimizing their free energy. My project is a continuation of previous work by my supervisor Francesco Sannino [arXiv:2507.06368v2], which lacks models including scalars. I first focus on possible chiral theories (Georgi–Glashow and Bars–Yankielowicz models), that includes scalars in different representations that can be made completely asymptotically free. I then derive the free energy for theories with chiral fermions and scalars, with the goal of minimizing it and adding the results to the atlas of possible GUTs that include scalars. In parallel with this work, I am also involved in a project on infinite heat order, a framework related to high-temperature behaviour and symmetry non-restoration, which builds on a previous paper by my supervisor [arXiv:2012.08428v2]. The focus is on models that exhibit symmetry non-restoration and are well defined in the UV.

Besides my master's thesis, I have completed two other projects during my master's degree. One project was on topological anomalies in quantum field theory, which I completed while on exchange. The project deepened my understanding of quantum field theory by studying the perturbative chiral anomaly and the non-perturbative Witten anomaly. The Fujikawa and triangle diagrams methods were used, and the chiral anomaly was considered topologically via the Atiyah-Singer index theorem. The Witten anomaly especially opened my eyes to how the topology of certain gauge theories restricts the allowed number of Weyl fermion doublets. This work highlighted how anomalies explain known phenomena, like neutral pion decay, and guide the development of consistent extensions of the Standard Model.

The second project was on percolation models as a comparison to SIR (susceptible, infected, recovered) models initiated at the beginning of the COVID-19 pandemic. My project was focused on treating infection rate as a phase transition and enhanced my programming skills and model building abilities.

My bachelor's thesis in the field of cosmology, examined whether the structure of the universe could explain its accelerated expansion. This work connects to a broader question on whether inhomogeneities can influence large-scale expansion. Throughout the project, I gained experience with deriving and manipulating complex dynamical equations, implementing numerical models, and analysing how local interactions can produce global effects.

During my bachelor's studies, I also completed several projects, which eventually led me to work on research involving dark matter and neural networks. Machine learning provides a non-parametric way to explore which density profiles are most consistent with observations. Inspired by my growing interest in machine learning and neural networks, I also volunteered to do lab work to support a research project on hyperspectral cameras and their use in optimizing wine production, using a neural network to analyse the hyperspectral data.

Across these projects, I have developed a strong foundation in theoretical physics, computational modelling, and data-driven analysis, and I am excited to continue exploring the fundamental structure of the universe through both analytical and computational methods within a theoretical framework.

## SCIENTIFIC INTERESTS

Please write a short summary of the area of physics that you find most interesting, and why. Do not exceed this page.

My scientific interests originate from a desire to understand the fundamental interactions that govern the smallest building blocks of the universe. I am drawn to questions that are simple to ask but immensely difficult to answer. Questions about why the forces of nature take their form and how particles acquire their properties. I am particularly fascinated by the role of symmetries in shaping physical laws, and how phenomena such as spontaneous symmetry breaking or anomalies determine particle behaviour. I wish to deepen my understanding of these principles and ultimately piece the clues together to answer: what symmetries govern physics beyond the Standard Model?

Even though the Standard Model is remarkably successful up to a few TeV, it leaves many questions unanswered, including the absence of gravity, the origin of baryon asymmetry, the replication of fermion families and their mixing, dark matter, and the hierarchy of masses. Studying the Higgs mechanism and flavour physics may provide hints about higher-energy dynamics. The interplay between low- and high-energy phenomena is particularly exciting to me, as it can reveal how subtle patterns at accessible energies reflect deeper organizing principles.

Flavour physics and the Higgs sector provide complementary windows into these questions. The replication of fermion families, the hierarchy of masses, and the structure of mixing matrices remain unexplained, suggesting the existence of deeper symmetries. The scalar nature of the Higgs allows it to probe the mechanisms behind mass generation. Taken together, these sectors point toward the symmetries and structures that shape particle masses, mixings, and interactions, suggesting possible approaches to extend the Standard Model.

Understanding these connections within the framework of quantum field theory, which unites symmetry principles and the dynamics of fundamental fields, sparked my interest from the first time I encountered it. This is why I chose to focus my projects and thesis on these topics. I yearn to use this framework to study the interplay of flavour, Higgs dynamics, and beyond-Standard-Model physics, with the hope of uncovering the deeper principles governing fundamental interactions. I am also motivated by open questions at the interface of quantum field theory and gravity, by the possibility of glimpsing quantum gravity through effective field theory methods embedded in the quantum field theory framework.

For my PhD, I hope to contribute to research in theoretical particle physics, especially in areas involving, Higgs and flavour dynamics, and the construction of consistent beyond-the-Standard-Model frameworks. Ultimately, I aim to help clarify the underlying principles that determine the structure of our universe.

## **HONORS, SCHOLARSHIPS, PRIZES AND AWARDS**

Please list any awards relevant to this application, giving dates and a brief description.

I applied for scholarships when going on exchange at University of Calgary. I received scholarships from the following foundations in 2024:

**Zonta Denmark** “Women in STEM, 2024”: given for remarkable achievements among women aged 18-35 in STEM.

**Niels Bohr Foundation:** given as a travel grant for students and young graduates within the fields of astronomy, mathematics, and physics.

**Doctor Else Mogensen Foundation:** given to significantly qualified, diligent and gifted students in natural sciences at University of Southern Denmark.

**William Demant Foundation:** given to selected students who are going on exchange with a high average.

**Knud Højgaard Foundation:** given to selected students who are going on exchange.

**Torben and Alice Frimodts Foundation:** given to students who are believed to support science, conservation or humanitarian causes.

## OUTSIDE INTERESTS

Please list your outside interests that are not directly related to your studies.

Outside of physics, I spend my time working, holding multiple jobs, most of which relate to outreach in physics. When I'm not working or studying, I play handball, a sport that has been an important part of my life since I was five. It provides me with a space to relax and recharge, and has also given me lasting friendships and a strong sense of determination.

## HOW DID YOU LEARN ABOUT US?

☒ Internet web site    ☐ Poster    ☐ Friends, colleagues, other students, faculty members

☐ Advertisement in scientific journals, please specify \_\_\_\_\_

☐ Other; please specify \_\_\_\_\_